

Lecture 11: Hypothesis Testing

CMSC/DATA 320: Introduction to Data Science

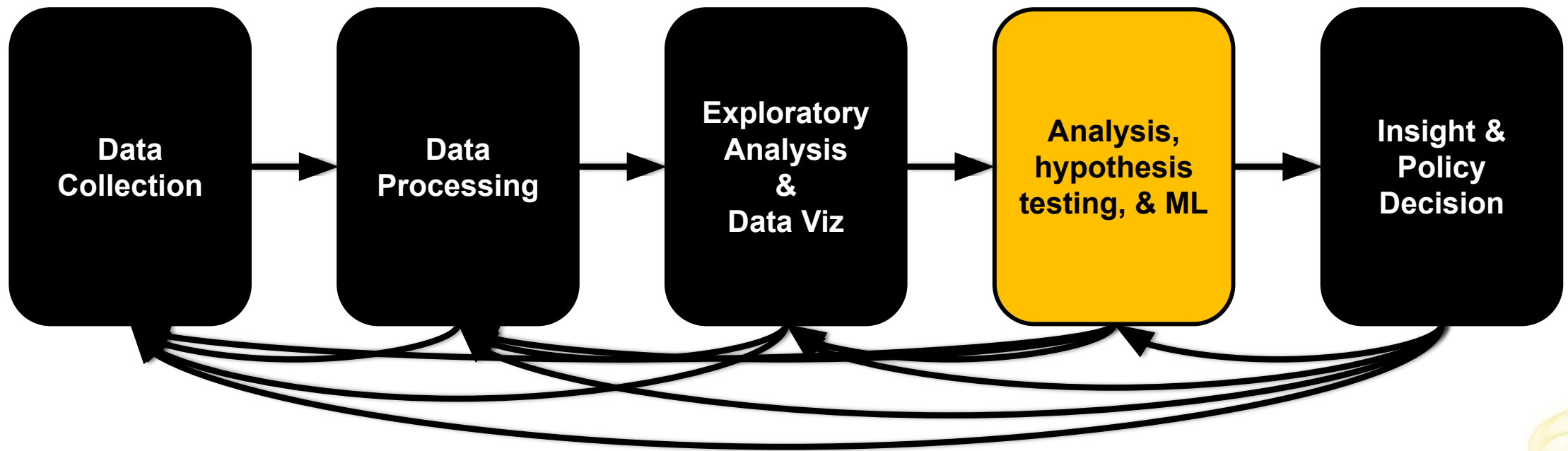
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Announcements

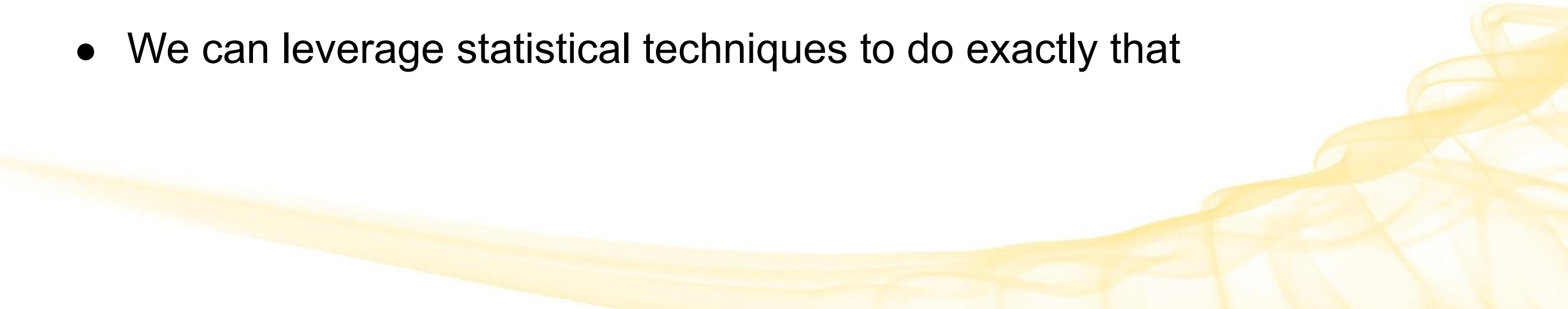
- PA2, WA2 grades out today
- Quiz 5 grades out today



Hypothesis Testing, Inference

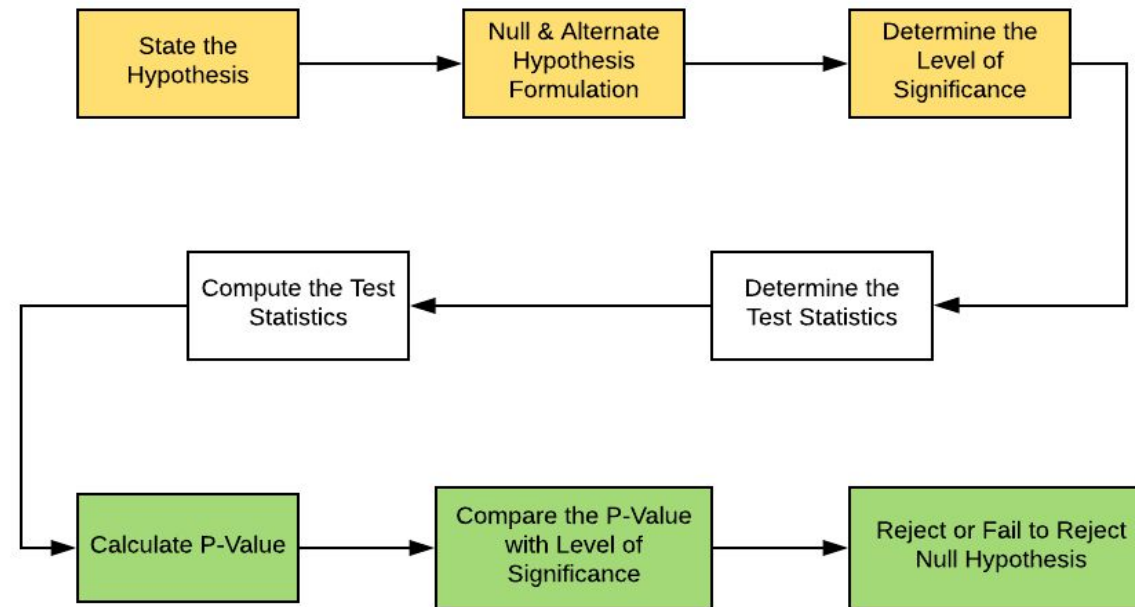


Hypothesis Testing – a statistical technique

- Previously, we went over a bunch of statistics and probability theories and formulae
 - Why did we do that?
 - We're going to use them!
 - Often we want to check whether a stated hypothesis is true or not
 - Or at least know how *likely* it is to be true or not
 - We can leverage statistical techniques to do exactly that
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Hypothesis Testing

Hypothesis testing is a statistical method used to make informed decisions or draw conclusions about a *population* based on a *sample of data*.



Hypothesis Testing Workflow

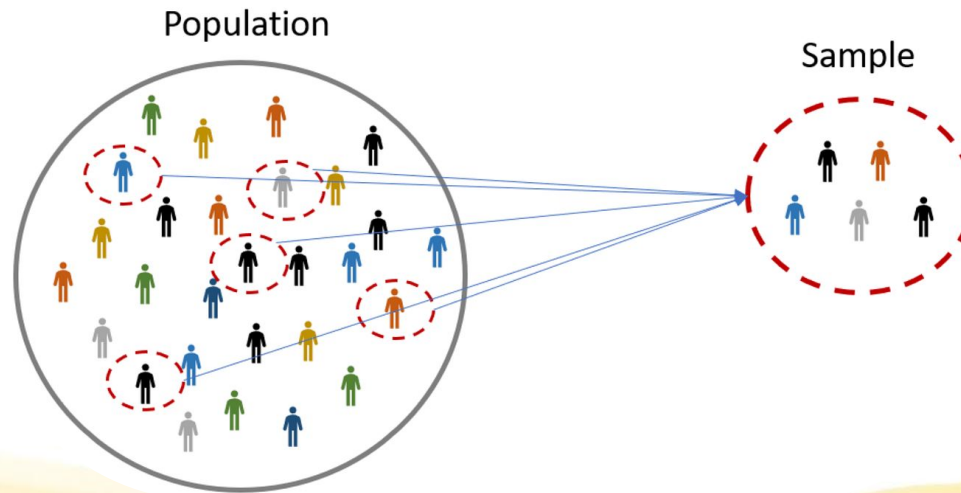
Hypothesis Testing

Hypothesis testing is a statistical method used to make informed decisions or draw conclusions about a **population** based on a **sample of data**.

RECAP:

Population: The **entire group** that you are interested in studying.

Sample: A **subset of the population** used for analysis.

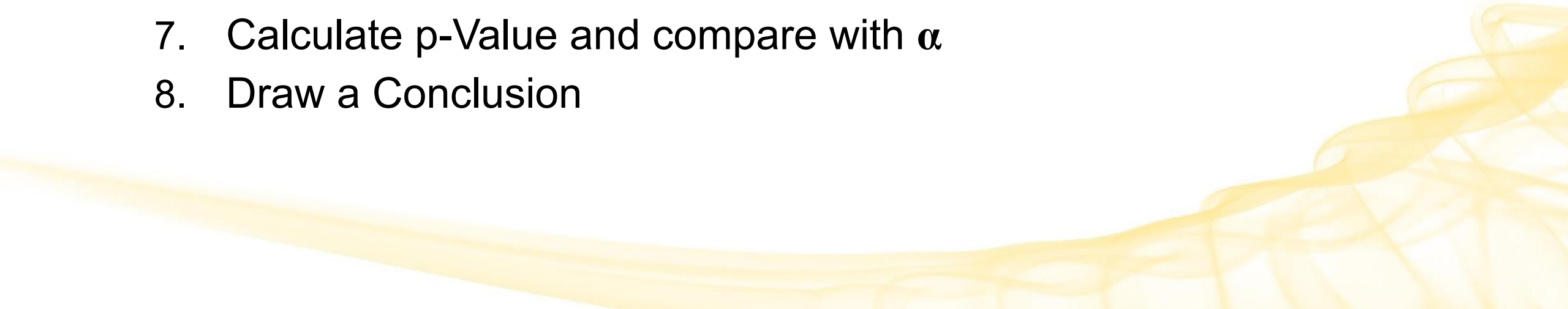


Statistical Inference

2 types of questions *we'll* ask:

- Is the mean of the 1 sample we have enough evidence to say something about a population mean?
 - Are the means of the 2 samples we have from the same population different enough?
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Main Steps of Hypothesis Testing

1. State the Null Hypothesis H_0
 2. State the Alternative Hypothesis H_a
 3. Pick a Level of Significance α
 4. Choose a Test
 5. Collect Data
 6. Calculate a test Statistic
 7. Calculate p-Value and compare with α
 8. Draw a Conclusion
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Create a Plan

Execute the Plan

Analyze the Results

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State / Formulate
The Hypothesis

Hypothesis

A **premise** or **claim** that we want to test/investigate.



Hypothesis

A **premise** or **claim** that we want to test/investigate.

- A statement about one or more populations.
- Usually concerned with the **parameters** (such as the **mean** (average), **variance** (spread), or **proportion** (percentage)) of the **population** (*help to understand the overall characteristics of the population*).

Ex. A hospital administrator may want to test the hypothesis that the *average length of stay* of patients admitted to the hospital is 5 days

Hypothesis Examples (for 1 sample testing)

- The average length of stay of patients admitted to the hospital is 5 days
 - The coin we're flipping is fair (50/50 chance of heads vs tails)
 - The average height of women in the United States is 5'5
 - Data Scientists prefer using Python over R
 - People are more likely to click on an ad if there is a picture of a kitten on it
 - US high school students spend 92¢ each per day on average on candy + soda
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There are two hypotheses in hypothesis testing

1. **Null Hypothesis:** The hypothesis to be tested
 - a. Def: A statement of **no effect** or **no difference** (currently established).
 - b. Represents currently accepted value for a parameter
 - c. Suggests that the observed data does not deviate from the population
 - d. Denoted as H_0
2. **Alternative Hypothesis** (a.k.a. Research Hypothesis): The statement you will adopt if the evidence (data) is strong enough to reject the null hypothesis.
 - a. Contradicts the null hypothesis.
 - b. Suggests that the observed data is **different** than the population
 - c. Represents what you think might be true
 - d. Denoted as H_a (or sometimes H_1)

*H_0 and H_a are **mutually exclusive**: only one can be true*

Example:

Let's say, **it is believed** that a candy making machine makes chocolate bars that are on average **5 grams in weight**. A worker **claims that** the machine after maintenance **no longer makes 5 gram bars**. Write down H_0 and H_a .

Null Hypothesis $H_0 : \mu = 5$ gram.

Alternative Hypothesis $H_a : \mu \neq 5$ gram.



H_0 and H_a are mathematical opposite (mutually exclusive)

Possible Outcomes of the Test

Let's say, it is believed that a candy making machine makes chocolate bars that are on average 5 grams in weight. A worker claims that the machine after maintenance no longer makes 5 gram bars. Write down H_0 and H_a .

$H_0: \mu = 5 \text{ grams.}$ $H_a: \mu \neq 5 \text{ grams.}$

This test can lead to TWO possible outcomes.

Possible Outcomes of this test:

1. **Reject** the null hypothesis
2. **Fail to Reject** the null hypothesis



Criteria: Based on evidence from sample data.

Possible Outcomes of the Test

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Q: How do you do the testing? How do you actually decide to reject the null or not?

Possible Outcomes of the Test

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A: Do Test Statistics

→ this is computed from sample data (during the execution of a statistical test)
used to decide possible outcome

Steps for a 1-sample test

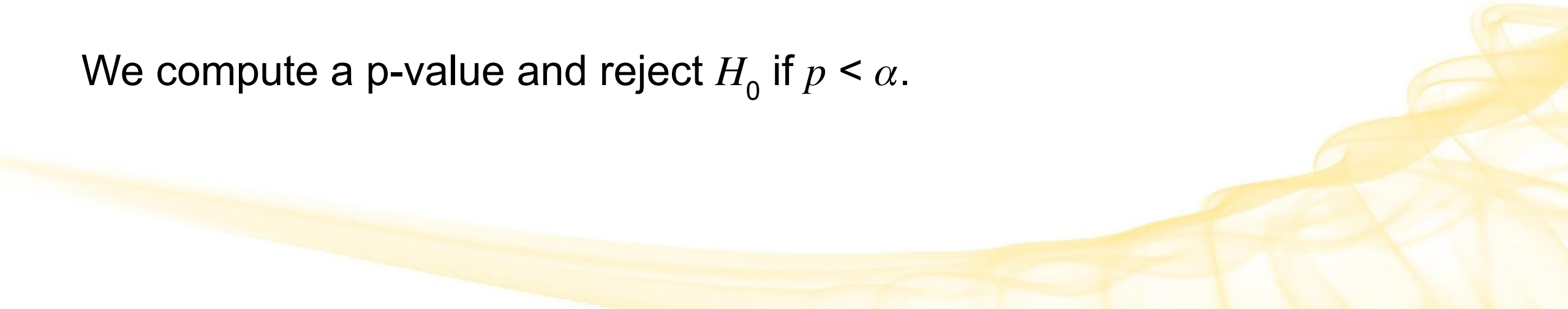
We want to test whether the mean (or proportion) of a population equals some claimed value.

We set up our hypotheses.

We calculate the t statistic:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

We compute a p-value and reject H_0 if $p < \alpha$.



Steps for a 2-sample test

We want to test whether the two samples are from the same population or not from the same population.

We set up our hypotheses.

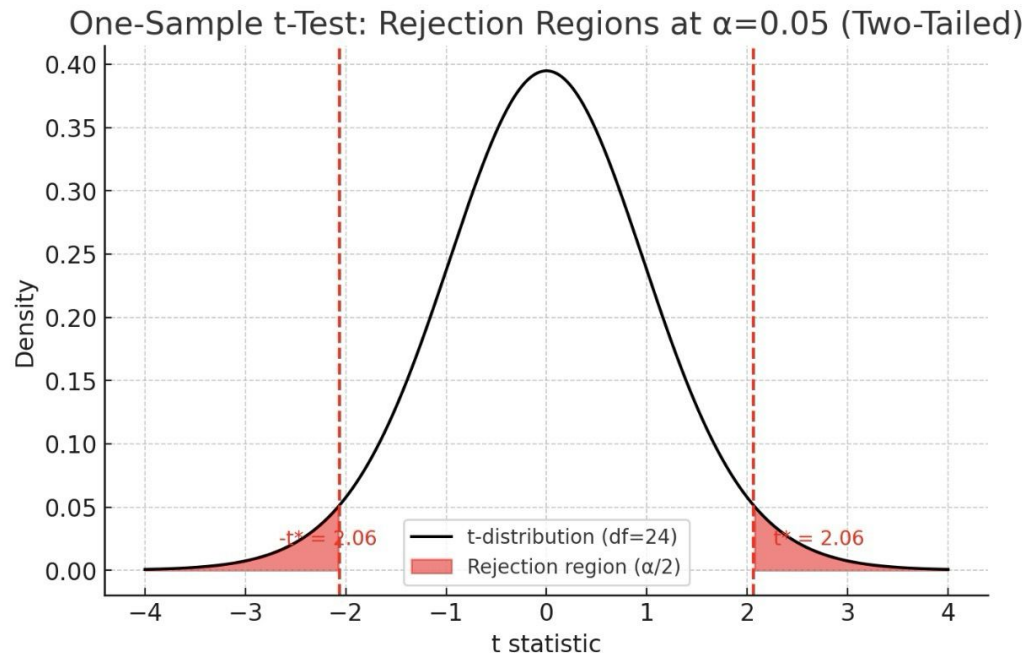
We calculate the t statistic and the associated p value (math is complicated).

We reject H_0 if $p < \alpha$.



T-statistic distribution and p values

We compute a p-value and reject H_0 if $p < \alpha$.



To reject or not to reject?

What step helps us **decide whether to reject** the null hypothesis or not, and **how confident** can we be in our decision?



Set the criteria for a decision:

Level of Significance (α)



Significance Level

Significance Level (usually denoted α): is a measure of the **strength of the evidence used as a threshold** (criterion) for deciding whether to reject or fail to reject the null hypothesis in a hypothesis test.

- α is not something we calculate. It's something we *choose*.
- It is our decided probability of rejecting the null hypothesis when it's actually true.
 - Choice of value is a tradeoff between sensitivity and false positives
 - Ex. How problematic would a false positive be?
 - Common α values are **5% or 1%** (typically $0 \leq \alpha \leq 0.05$)

Significance level helps you determine
how confident you need to be in your results
before you make a decision.

Why do we need α ?

Hypothesis testing is an *inferential* procedure.

- It will determine whether your sample evidence is strong enough to suggest that the effect/relationship exists in the wider population.

But what if there is a sampling error?

- Sample data = evidence
- α = how strong the sample evidence must be before deciding if the results are statistically significant.
 - The maximum risk of making a false positive that you are willing to accept

Why do we set α ahead of time?

To make sure that we are setting an **unbiased** threshold for decision-making.

If we were to wait until after the results, might we be (knowingly or unknowingly) biased in our choice of value?



Significance Level Example

Suppose you're testing a new treatment for headaches.

Your null hypothesis (H_0) : **the treatment has no effect**

Your alternative hypothesis (H_a) : **the treatment reduces headaches**

- You set your significance level (α) at 0.05, meaning **you're willing to accept a 5% chance of mistakenly** concluding that the treatment works when it actually doesn't.

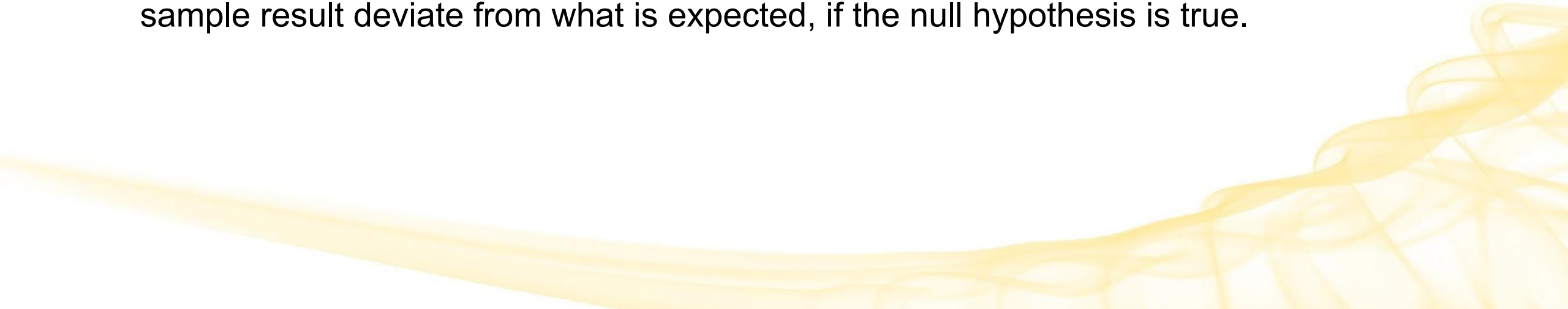
Statistical Tests & Test Statistics



Test Statistic

Definition: A test statistic is a **numerical value** calculated from **sample data** during the hypothesis testing process.

A test statistic:

- **Quantifies the evidence** from the data against the null hypothesis and helps determine whether to reject or fail to reject the null hypothesis.
 - **Measures how close** the sample results align with the null hypothesis or how much the sample result deviate from what is expected, if the null hypothesis is true.
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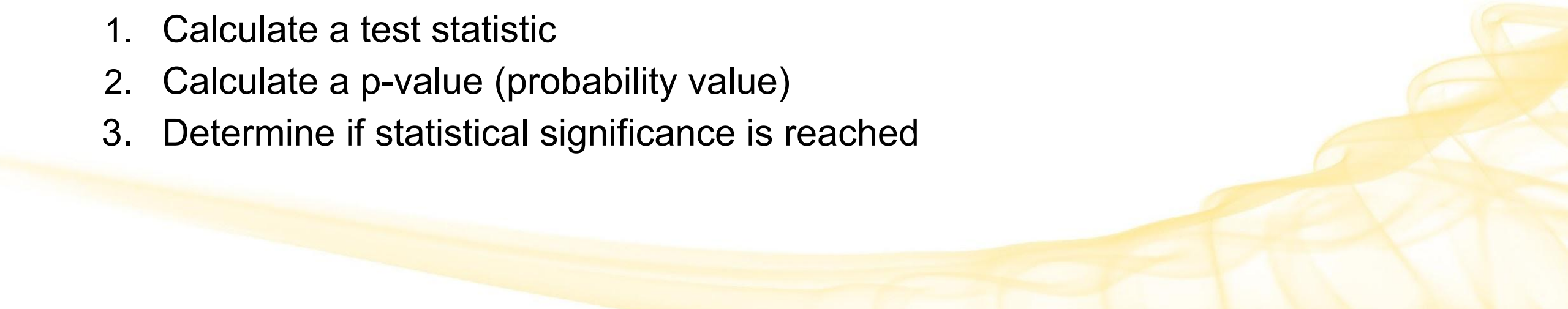
*Tells you if your data **agrees** or **disagrees** with your starting idea (the null hypothesis). It shows how strong your evidence is either for or against your idea.*

Statistical Test

A **statistical test** refers to the **overall method** or procedure used to decide whether to reject the null hypothesis.

It is a method of statistical inference used to decide whether the data at hand sufficiently support a particular hypothesis.

Statistical tests:

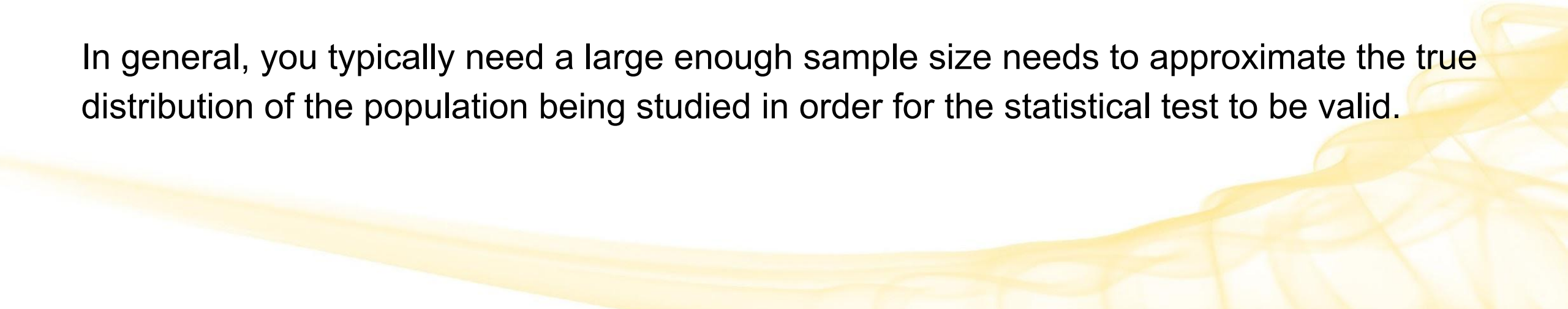
1. Calculate a test statistic
 2. Calculate a p-value (probability value)
 3. Determine if statistical significance is reached
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Choosing a Statistical Test

The **choice of the appropriate statistical test** depends on various factors, such as:

- ❑ The research question
- ❑ The specific hypothesis being tested
- ❑ The type of data (ex. categorical or numerical, continuous or discrete)
- ❑ The number of groups being compared

In general, you typically need a large enough sample size needs to approximate the true distribution of the population being studied in order for the statistical test to be valid.



Types of Statistical Tests

- **Parametric**: Precise requirements with strong inference.
 - **Regression** (*future topic*): tests cause-and-effect relationships
 - **Comparison**: tests differences between (compares) group means
 - **T-test**: compare the means of 2 groups. Very common.
 - **ANOVA** (Analysis of Variance): compare means of > 2 groups
 - **Z-test**: compares against the population mean (need large n and known σ)
 - **Correlation**
 - **Pearson Correlation Coefficient**: tests linear correlation between 2 variables (how related)
- **Non-parametric**: No assumptions about the distribution (ex. normality).
Not as accurate as parametric but useful when common statistical assumptions are violated
 - **Chi-Squared Test**: compares two categorical variables

What are common statistical assumptions?

1. **Independent observations**
 - Observations in your test are not related/correlated
 2. **Homogeneous variables**
 - Groups being compared have similar variances
 3. **Normal data**
 - The data follows a normal distribution
 4. **Random sampling**
- 

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If you know these are violated, try a non-parametric test like Chi-Squared

T-test for 1 sample

Determine if the mean of a **single sample** is significantly different from a **known** (or hypothesized) **population mean**.

- Commonly used when you've collected sample data from a single group and want to compare its mean to a specific value or a hypothesized population mean.

T-test for 2 samples

The two sample t-test (also referred to as the unpaired, or independent, t-test) is **used to compare the means of two different samples.**

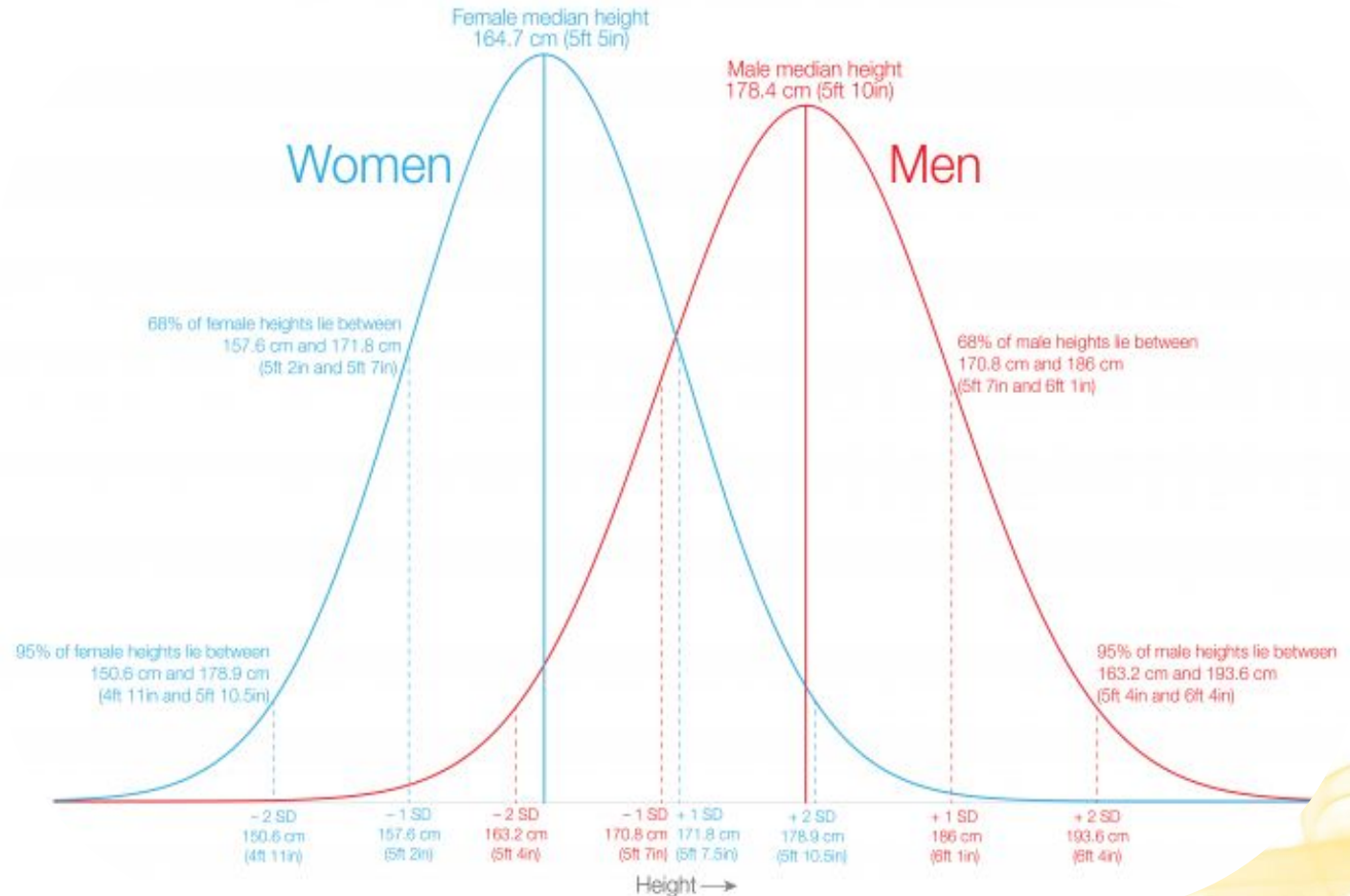
Example: We have noticed most humans fall into one of two distinct categories – male or female. **We would like to know if our sample of males is taller than our sample of females.**

Can we just take the average of the two samples?

NO. Simply taking the average of the two samples is not sufficient to determine if one group is taller than the other → it does not account for variability within each group.

Two-Sample t-test

Instead, we conduct a statistical test to determine whether there is a statistically significant difference between the two groups.



When can we use the T-test?

We can use t-test for **continuous data** obtained from a random sample that follows a normal distribution.

Assumptions: For the t-test, we assume:

- The data is **normally distributed**
- We care about **the mean** → For example, whether there is a significant difference in the means of the two groups.

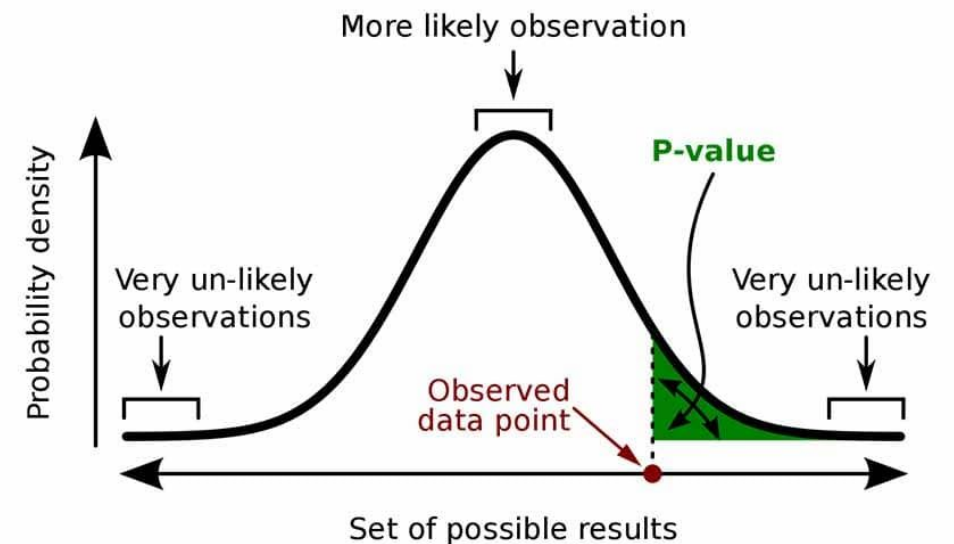
Q: What if my data isn't nearly normally distributed?

- If your sample sizes are small, you might not be able to test for normality. You might need to rely on your understanding of the data/context.
- When you cannot safely assume normality, you can **perform a nonparametric test** that doesn't assume normality.

p-value of a statistical test

A measure used to quantify the strength of evidence **against** the null hypothesis.

- ❑ Represents the probability of observing the test statistic, or a more extreme value, *if the null hypothesis is true*.
 - $p\text{-value} = P(\text{statistic} \mid H_0 = \text{true})$
- ❑ A low p-value indicates that the observed data are **unlikely** under the null hypothesis, suggesting strong evidence against it.
- ❑ If the $p\text{-value} \leq \alpha$, the null hypothesis is rejected.



A **p-value** (shaded green area) is the probability of an observed (or more extreme) result assuming that the null hypothesis is true.